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Inventor(s)	Mitchell; James D. et al.

VASCULAR ACCESS DEVICES, SYSTEMS, AND METHODS

Abstract

Vascular access devices and associated systems and methods are disclosed herein. According to some embodiments, the present technology includes a method of treatment comprising implanting a vascular access device through an incision at a first location and implanting an electronic device through the incision at a second location, the second location being different than the first location.

Inventors:	**Mitchell; James D. (Walnut Creek, CA), Thoreson; Andrew A. (Orono, MN), Aklog; Lishan (Purchase, NY), Deguzman; Brian J. (Paradise Valley, AZ), O'Neill; Stephen J. (Waltham, MA)**
Applicant:	**Veris Health Inc** (New York, NY)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] The present application claims the benefit of priority to U.S. Provisional Application No. 63/363,731, filed Apr. 28, 2022.

TECHNICAL FIELD
[0002] The present technology relates to implantable medical devices and associated systems and methods of use. Particular embodiments of the present technology are directed to vascular access devices, systems, and methods.
BACKGROUND
[0003] Vascular access devices (e.g., vascular access ports) are minimally invasive, surgically implanted devices that provide relatively quick and easy access to a patient's central venous system for the purpose of administering intravenous medications, such as chemotherapeutic agents. Conventional vascular access devices are commonly used for patients requiring frequent, repeated intravenous administration of therapeutic agents or fluid, repeated blood draws, and/or for patients with difficult vascular access.
[0004] Vascular access devices such as vascular access ports typically include a reservoir attached to a catheter. The entire unit is placed completely within a patient's body using a minimally invasive surgical procedure. In most cases, the reservoir is placed in a small pocket created in the upper chest wall just inferior to the clavicle, and the catheter is inserted into the internal jugular vein or the subclavian vein with the tip resting in the superior vena cava or the right atrium. However, such vascular access devices can be placed in other parts of the body and/or with the catheter positioned in alternative sites as well. In conventional devices, the reservoir is typically bulky such that the overlying skin protrudes, allowing a clinician to use palpation to localize the device for access when it is to be used for a medication infusion or aspiration of blood for testing. A self-sealing cover (e.g., a thick silicone membrane) is disposed over and seals the reservoir, allowing for repeated access using a non-coring (e.g., Huber type) needle that is inserted through the skin and into the port. This access procedure establishes a system in which there is fluid communication between the needle, the vascular access device, the catheter, and the vascular space, thereby enabling infusion of medication or aspiration of blood via a transcutaneous needle.
[0005] Conventional vascular access devices are bulky by design to allow a clinician to localize the device by palpation. To be accurately accessed by a clinician, the vascular access device needs to be either visualized or palpated under the skin. Additionally, conventional vascular access ports have no electronic components and no internal power source. Accordingly, there is a need for improved vascular access devices.
SUMMARY
[0006] The subject technology is illustrated, for example, according to various aspects described below, including with reference to FIGS. **1-9**. Various examples of aspects of the subject technology are described as numbered clauses (1, 2, 3, etc.) for convenience. These are provided as examples and do not limit the subject technology. [0007] 1. A method of treatment, comprising: [0008] implanting a vascular access device through an incision at a first location; and [0009] implanting an electronic device through the incision at a second location, the second location being

different than the first location. [0010] 2. A method of treatment, comprising: [0011] securing a vascular access device to an electronic device via a coupler; and [0012] implanting the vascular access device, the electronic device, and the coupler together through an incision. [0013] 3. A method of treatment, comprising: [0014] implanting a vascular access device; and [0015] implanting an electronic device after implanting the vascular access device; and [0016] after implanting the electronic device, securing the vascular access device and the electronic device to a coupler. [0017] 4. A method of treatment, comprising: [0018] implanting an electronic device; and [0019] implanting a vascular access device after implanting the electronic device; and [0020] after implanting the vascular access device, securing the vascular access device and the electronic device to a coupler. [0021] 5. A method of treatment, comprising: [0022] securing a vascular access device to an electronic device by engaging a coupling portion of the vascular access device with a coupling portion of the electronic device; and [0023] implanting the vascular access device and the electronic device together through an incision. [0024] 6. A method of treatment, comprising: [0025] implanting a vascular access device; and [0026] implanting an electronic device after implanting the vascular access device; and [0027] after implanting the electronic device, securing the vascular access device to the electronic device by engaging a coupling portion of the vascular access device with a coupling portion of the electronic device. [0028] 7. A method of treatment, comprising: [0029] implanting an electronic device; and [0030] implanting a vascular access device after implanting the electronic device; and [0031] after implanting the vascular access device, securing the vascular access device to the electronic device by engaging a coupling portion of the vascular access device with a coupling portion of the electronic device. [0032] 8. The method of any one of Clauses 1 to 5, wherein the vascular access device comprises: [0033] a hub comprising a housing defining a reservoir configured to receive a fluid therein; and [0034] a catheter comprising a proximal end portion configured to be secured to the hub, a distal end portion configured to be positioned within a body lumen of a patient, and a lumen extending therethrough, the lumen being in fluid communication with the reservoir. [0035] 9. The method of any one of Clauses 1 to 8, wherein the electronic device comprises: [0036] a sensing element, the sensing element being configured to obtain data characterizing a physiological parameter of a patient. [0037] 10. The method of Clause 9, wherein the electronic device comprises: [0038] a data communications module communicatively coupled to the sensing element and configured to transmit the data obtained by the sensing element to an external computing device. [0039] 11. The method of any one of Clauses 8 to 10, wherein the body lumen is a blood vessel. [0040] 12. The method of any one of Clauses 8 to 10, wherein the body lumen is a chamber or valve of the heart. [0041] 13. The method of any one of Clauses 8 to 10, wherein the body lumen is a peritoneal space, a renal pelvis or ureter, a bladder, a gallbladder, a stomach, a small bowel, a large bowel, a lung, a pleural space, or a pericardial space. [0042] 14. The method of any one of Clauses 2 to 4, wherein the coupler includes a first receiving portion configured to be secured to the vascular access device and a second receiving portion configured to be secured to the electronic device. [0043] 15. The method of Clause 1, wherein the first and second locations are within the same subcutaneous pocket.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0044] Many aspects of the present disclosure can be better understood with reference to the following drawings. The components in the drawings are not necessarily to scale. Instead, emphasis is placed on illustrating clearly the principles of the present disclosure.

[0045] FIG. **1** is a schematic representation of a system for monitoring the health of a patient via an implanted medical system in accordance with the present technology.

[0046] FIG. **2** shows a vascular access device configured for use with the system of FIG. **1** and in

accordance with several embodiments of the present technology.
[0047] FIG. **3** shows an electronic device configured for use with the system of FIG. **1** and in accordance with several embodiments of the present technology.
[0048] FIGS. **4**A and **4**B show a vascular access device and a coupler, respectively, configured for use with one another and in accordance with several embodiments of the present technology.
[0049] FIG. **4**C shows the coupler of FIG. **4**B coupled to a vascular access device and an electronic device in accordance with several embodiments of the present technology.
[0050] FIGS. **5**A and **5**B show a vascular access device and a coupler, respectively, configured for use with one another and in accordance with several embodiments of the present technology.
[0051] FIGS. **6**A and **6**B show a vascular access device and a coupler, respectively, configured for use with one another and in accordance with several embodiments of the present technology.
[0052] FIGS. **7**A and **7**B show a vascular access device and a coupler, respectively, configured for use with one another and in accordance with several embodiments of the present technology.
[0053] FIG. **8** shows a side cross-sectional view of a receiving portion of a coupler configured in accordance with several embodiments of the present technology.
[0054] FIG. **9** shows a coupler configured in accordance with several embodiments of the present technology.
[0055] FIGS. **10**A-**10**D show a system configured in accordance with several embodiments of the present technology. FIGS. **10**A-**10**C show the vascular access device and the electronic device coupled to the coupler. FIG. **10**D is an isolated view of the coupler.
[0056] FIGS. **11**A and **11**B show a system in an assembled state and configured in accordance with several embodiments of the present technology.
[0057] FIG. **11**C is an isolated view of the electronic device of the system shown in FIGS. **11**A and **11**B.
[0058] FIGS. **12**A and **12**B show a system in an assembled state and configured in accordance with several embodiments of the present technology.
[0059] FIG. **12**C is an isolated view of the electronic device of the system shown in FIGS. **12**A and **12**B.
[0060] FIGS. **13**A and **13**B show a system in an assembled state and configured in accordance with several embodiments of the present technology.
[0061] FIG. **14** illustrates a system in an assembled state and configured in accordance with several embodiments of the present technology.
[0062] FIGS. **15**A and **15**B show a system in an assembled state and configured in accordance with several embodiments of the present technology.
[0063] FIGS. **16**A and **16**B show a system in an assembled state and configured in accordance with several embodiments of the present technology.
[0064] FIG. **16**C is an isolated view of the electronic device of the system shown in FIGS. **16**A and **16**B.
[0065] FIGS. **17**A-**17**C show a system in various configurations in accordance with several embodiments of the present technology.
[0066] FIG. **18** illustrates a system in as assembled state and configured in accordance with several embodiments of the present technology.
[0067] FIGS. **19**A and **19**B show a system in an assembly state and configured in accordance with several embodiments of the present technology.

DETAILED DESCRIPTION

[0068] The present technology relates to implantable medical devices such as vascular access devices and associated systems and methods of use. Specific details of several embodiments of the technology are described below with reference to FIGS. **1-9**.
[0069] The devices and systems of the present technology may be equipped with electronic components that provide a platform for remote monitoring of the device and/or patient. For

example, several of the systems disclosed herein include an implantable vascular access device and a separate, implantable electronic device. The electronic device can include a sensing element that is configured to obtain patient physiological data while the electronic device is implanted within the patient, and determine one or more physiological parameters based on the data. The system may determine certain physiological parameters, for example, that indicate one or more symptoms of a medical condition that requires immediate medical attention or hospitalization. Such physiological parameters can include those related to temperature, patient movement/activity level, heart rate, respiratory rate, blood oxygen saturation, and/or other suitable parameters described herein. The vascular access device and the electronic device may be implanted at the same time and through the same incision, which is convenient for the physician and procedurally cost effective. In some embodiments the vascular access device and electronic device are coupled to one another prior to implantation (i.e., ex vivo). According to some embodiments, the vascular access device and electronic device are coupled to one another in vivo.

[0070] According to various embodiments of the present technology, the vascular access device comprises a fluid reservoir and a catheter configured to be fluidically coupled to the reservoir. The vascular access device is configured to be implanted in a subcutaneous pocket at an accessible location (such as the chest), and the catheter can extend from the reservoir to a remote vascular location to provide convenient, repeatable access to the patient's venous or arterial system. In some embodiments, the catheter can provide access to a body cavity or hollow organ such as the peritoneal space, the renal pelvis or ureter, the bladder, the gallbladder, the stomach, the small bowel, the large bowel, the lung, the pleural space, or the pericardial space.

[0071] FIG. **1** is a schematic representation of a system **10** for remote monitoring and vascular access. The system **10** comprises a modular implantable system **100** in accordance with the present technology. The implantable system **100** is configured to be implanted within a human patient H at an accessible location, such as at a subcutaneous location along an upper region of the patient's chest. For example, according to several embodiments, the implantable system **100** can be implanted in a subcutaneous pocket created in the patient's upper chest wall, just inferior to the clavicle. As shown in FIG. **1**, the implantable system **100** may include a vascular access device **101** and an electronic device **102**. The vascular access device **101** can comprise a reservoir configured to receive a therapeutic agent for intravascular administration, and a catheter extending from the reservoir and configured to be inserted into the vasculature at a remote location. The vascular access device **101** and the electronic device **102** can be implanted in the same or different locations and at the same time or at different times, as discussed in greater detail below.

[0072] The electronic device **102** can include electronic components that are configured to obtain data characterizing a physiological parameter of the patient and communicate that data to the system **10**. In some embodiments, the electronic device **102** includes a sensing element that is configured to obtain physiological measurements that are used by the system **10** to determine one or more physiological parameters indicative of the patient's health. As used herein, “sensing element” can refer to a single sensor or multiple sensors. The system **10** may detect a medical condition or associated symptom(s) based on the physiological parameter(s) and, optionally, provide an indication of the detected condition to the patient, caregiver, and/or medical care team.

[0073] As shown schematically in FIG. **1**, the electronic device **102** may be configured to communicate wirelessly with a local computing device **150**, which can be, for example, a smart device (e.g., a smartphone, a tablet, or other handheld device having a processor and memory), a special-purpose interrogation device, or other suitable device. Communication between the implantable system **100** and the local computing device **150** can be mediated by, for example, near-field communication (NFC), infrared wireless, Bluetooth, ZigBee, Wi-Fi, inductive coupling, capacitive coupling, or any other suitable wireless communication link. The implantable system **100** may transmit data including, for example, measurements obtained via the sensing element characterizing physiological parameters of the patient, patient medical records, device performance metrics (e.g., battery level, error logs, etc.), or any other such data obtained by or stored by the implantable system **100**. In some embodiments, the transmitted data is encrypted or otherwise obfuscated to maintain security during transmission to the local computing device **150**. The local computing device **150** may also provide instructions to the electronic device **102**, for example to obtain certain physiological measurements via the sensing element, to emit a localization signal, or to perform other functions. In some embodiments, the local computing device **150** may be configured to wirelessly recharge a battery of the implantable system **100**, for example via inductive charging.

[0074] The system **10** may further include first remote computing device(s) **160** (or server(s)), and the local computing device **150** may in turn be in communication with first remote computing device(s) **160** over a wired or wireless communications link (e.g., the Internet, public and private intranet, a local or extended Wi-Fi network, cell towers, the plain old telephone system (POTS), etc.). The first remote computing device(s) **160** may include one or more own processor(s) and memory. The memory may be a tangible, non-transitory computer-readable medium configured to store instructions executable by the processor(s). The memory may also be configured to function as a remote database, i.e., the memory may be configured to permanently or temporarily store data received from the local computing device **150** (such as one or more physiological measurements or parameters and/or other patient information).

[0075] In some embodiments, the first remote computing device(s) **160** can additionally or alternatively include, for example, server computers associated with a hospital, a medical provider, medical records database, insurance company, or other entity charged with securely storing patient data and/or device data. At a remote location **170** (e.g., a hospital, clinic, insurance office, medical records database, operator's home, etc.), an operator may access the data via a second remote computing device **172**, which can be, for example a personal computer, smart device (e.g., a smartphone, a tablet, or other handheld device having a processor and memory), or other suitable device. The operator may access the data, for example, via a web-based application. In some embodiments, the obfuscated data provided by the implantable system **100** can be de-obfuscated (e.g., unencrypted) at the remote location **170**.

[0076] In some embodiments, the electronic device **102** may communicate with remote computing devices **160** and/or **172** without the intermediation of the local computing device **150**. For example, the electronic device **102** may be connected via Wi-Fi or other wireless communications link to a network such as the Internet. In other embodiments, the electronic device **102** may be in communication only with the local computing device **150**, which in turn is in communication with remote computing devices **160** and/or **172**.

[0077] FIG. **2** shows an example of a vascular access device **200** (or “device **200**”) configured for use with the systems **10** and implantable systems **100** of the present technology. As shown in FIG. **2**, the device **200** comprises a hub **202** and a catheter **220** configured to be permanently or detachably coupled to the hub **202**. The hub **202** may comprise a housing **208**, a fluid reservoir **210** contained within the housing **208**, and a septum **212** adjacent the reservoir **210** that is configured to receive a needle therethrough for delivery of a fluid (such as a therapeutic or diagnostic agent) to the reservoir **210**. The housing **208** may be made of a biocompatible plastic, metal, ceramic, medical grade silicone, or other material that provides sufficient rigidity and strength to prevent inadvertent needle puncture through the housing **208**. The septum **212** can be, for example, a self-sealing membrane made of silicone or other deformable, self-sealing, biocompatible material.

[0078] The catheter **220** can be configured to permanently or detachably couple to the hub **202** to be placed into fluid communication with the reservoir **210**. For example, as shown in FIG. **2**, the catheter **220** can have a proximal end portion **220*a*** configured to mate with an outlet port **214** of the hub **202** (e.g., via a barb connector or other suitable mechanical connection) and a distal end portion **220*b*** configured to be positioned within a blood vessel or a heart of a patient. The device **200** and/or implantable system **100** can comprise a single catheter **220** (as shown in FIG. **2**) or multiple catheters **220**. Moreover, the catheter **220** can define a single lumen or multiple lumens.

[0079] FIG. **3** schematically depicts an electronic device **300** (or "device **300**") configured for use with the systems **10** and implantable systems **100** of the present technology. The electronic device **300** is configured to obtain measurements indicative of a health of the patient as well as communicate with the system **100**. The electronic device **300** may include a sensing element **302**, a controller **304**, and a communications unit **306**, all carried by a housing **301**. The housing **301**, for example, can comprise a hermetically sealed enclosure. In some embodiments, the sensing element **302**, controller **304**, and/or communications unit **306** are carried by a printed circuit board (PCB). As discussed in greater detail below, the sensing element **302** can be configured to obtain data characterizing an identity parameter of the patient, a physiological parameter of the patient. As used herein, the term "sensing element" may refer to a single sensor or a plurality of discrete, separate sensors. Although one sensing element **302** is illustrated for clarity in FIG. **3**, in various embodiments, the device **300** may include more than two sensing elements **302** (e.g., two sensing elements, three sensing elements, four sensing elements, etc.).

[0080] As shown in FIG. **3**, in some embodiments the device **300** includes a power source **308**, such as a battery. In several of such embodiments, the device **300** can include an antenna (e.g., a coil) (not shown) that is configured to be inductively coupled to an external antenna (e.g., a coil). The electrical energy generated in the device antenna by the external antenna can be used to recharge and/or power the power source **308**. In certain embodiments, the device **300** does not include a power source **308** and only includes the device antenna. In such embodiments, the device **300** may only be powered when in the presence of the external antenna. In any case, the external antenna can be, for example, an interrogation device (e.g., local device **150** or another suitable device) or other device.

[0081] As shown in FIG. **3**, the electronic device **300** can have an elongated shape with a slim profile for improving patient comfort. In other embodiments, the electronic device **300** can have other shapes and configurations.

[0082] The controller **304** may include one or more processors, software components, and/or memory (not shown). In some examples, the one or more processors include one or more computing components configured to process measurements received from the sensing element **302** according to instructions stored in the memory. The memory may be a tangible, non-transitory computer-readable medium configured to store instructions executable by the one or more processors. For instance, the memory may be data storage that can be loaded with one or more of the software components executable by the one or more processors to achieve certain functions. In some examples, the functions may involve causing the sensing element **302** to obtain one or more measurements, such as data characterizing a physiological parameter of the patient. In another example, the functions may involve processing the data to determine one or more parameters and/or provide an indication to the patient and/or clinician of a health of the patient. For example, the functions may involve providing an indication to the patient and/or clinician of one or more symptoms or medical conditions associated with determined physiological parameters.

[0083] The communications unit **306** can be configured to securely transmit data between the device **300** and external computing devices (e.g., local computing device **150**, remote computing devices **160** and **172**, etc.). In some embodiments, the controller **304** includes a localization unit configured to emit a localization signal (e.g., lights that transilluminate a patient's skin, vibration, a magnetic field, etc.) to aid a clinician in localizing the device **300** when implanted within a patient.

[0084] The system **10** may be configured to continuously and/or periodically obtain measurements via the sensing element **302**.

[0085] In some embodiments, the sensing element **302** is built into the housing **301** such that only a portion of the sensing element **302** is exposed to the local physiological environment when the device **300** is implanted. For example, the sensing element **302** may comprise one or more electrodes having an external portion positioned at an exterior surface of the housing **301** and an internal portion positioned within the housing **301** and, optionally, wired to the controller **304**. In some embodiments the sensing element **302** may be completely contained within the housing **301**. For example, the sensing element **302** may comprise one or more pulse oximeters enclosed by the housing **301** and positioned adjacent a window in the housing **301** through which light emitted from the pulse oximeter may pass to an external location, and back through which light reflected from the external location may pass for detection by a photodiode of the pulse oximeter. In such embodiments the window may be, for example, a sapphire window that is brazed into place within an exterior wall of the housing **301**.

[0086] The sensing element **302** may comprise at least one sensor completely enclosed by the housing **301** and at least one sensor that is partially or completely positioned at an external location, whether directly on the housing **301** or separated from the housing **301** (but still physically coupled to the housing **301** via a wired connection, for example).

[0087] In some embodiments, the controller **304** processes at least some of the measurements to determine one or more parameters, and then transmits those parameters to one or more of the local computing device **150**, remote computing devices **160**, and/or remote computing devices **172** (with or without the underlying data). In some examples, the controller **304** may only partially process at least some of the measurements before transmitting the data to the local computing device **150**, remote computing devices **160**, and/or remote computing devices **172**. In such embodiments, the controller **304** may further process the received data to determine one or more parameters. The local computing device **150**, the remote computing devices **160**, and/or the remote computing devices **172** may also process some or all of the measurements obtained by the sensing element **302** and/or parameters determined by the sensing element **302** and/or the controller **304**.

[0088] In some embodiments, the sensing element(s) **302** and/or controller **304** may be configured to detect, identify, monitor, and/or communicate information by electromagnetic, acoustic, motion, optical, thermal, or biochemical sensing elements or means. The sensing element(s) **302** may include, for example, one or more temperature sensing elements (e.g., one or more thermocouples, one or more digital temperature sensors, one or more thermistors or other type of resistance temperature detector, etc.), one or more impedance sensing elements (e.g., one or more electrodes), one or more pressure sensing elements, one or more optical sensing elements, one or more flow sensing elements (e.g., a Doppler velocity sensing element, an ultrasonic flow meter, etc.), one or more ultrasonic sensing elements, one or more photoplethysmography (PPG) sensing elements (e.g., pulse oximeters, etc.), one or more chemical sensing elements, one or more movement sensing elements (e.g., one or more accelerometers), one or more pH sensing elements, an electrocardiogram ("ECG" or "EKG") unit, one or more electrochemical sensing elements, one or more hemodynamic sensing elements, and/or other suitable sensing devices.

[0089] The sensing element **302** may comprise one or more electromagnetic sensing elements configured to measure and/or detect, for example, impedance, voltage, current, or magnetic field sensing capability with a wire, wires, wire bundle, magnetic node, and/or array of nodes. The sensing element **302** may comprise one or more acoustic sensing elements configured to measure and/or detect, for example, sound frequency, within human auditory range or below or above frequencies of human auditory range, beat or pulse pattern, tonal pitch melody, and/or song. The sensing element **302** may comprise one or more motion sensing elements configured to measure and/or detect, for example, vibration, movement pulse, pattern or rhythm of movement, intensity of movement, and/or speed of movement. Motion communication may occur by a recognizable response to a signal. This response may be by vibration, pulse, movement pattern, direction, acceleration, or rate of movement. Motion communication may also be by lack of response, in which case a physical signal, vibration, or bump to the environment yields a motion response in the surrounding tissue that can be distinguished from the motion response of the sensing element **302**. Motion communication may also be by characteristic input signal and responding resonance. The sensing element **302** may comprise one or more optical sensing elements which may include, for example, illuminating light wavelength, light intensity, on/off light pulse frequency, on/off light pulse pattern, passive glow or active glow when illuminated with special light such as UV or “black light”, or display of recognizable shapes or characters. It also includes characterization by spectroscopy, interferometry, response to infrared illumination, and/or optical coherence tomography. The sensing element **302** may comprise one or more thermal sensing elements configured to measure and/or detect, for example, the temperature of the device **300** relative to surrounding environment, the temperature of the device **300** (or portion thereof), the temperature of the environment surrounding the device **300** and/or sensing element **302**, or differential rate of the device temperature change relative to surroundings when the device environment is heated or cooled by external means. The sensing element **302** may comprise one or more biochemical devices which may include, for example, the use of a catheter, a tubule, wicking paper, or wicking fiber to enable micro-fluidic transport of bodily fluid for sensing of protein, RNA, DNA, antigen, and/or virus with a micro-array chip.

[0090] In some aspects of the technology, the controller **304** and/or sensing element **302** may be configured to detect and/or measure the concentration of blood constituents, such as sodium, potassium, chloride, bicarbonate, creatinine, blood urea nitrogen, calcium, magnesium, and phosphorus. The system **10** and/or the sensing element **302** may be configured to evaluate liver function (e.g., by evaluation and/or detection of AST, ALT, alkaline phosphatase, gamma glutamyl transferase, troponin, etc.), heart function (e.g., by evaluation and/or detection of troponin, brain natriuretic peptide (BNP)), coagulation (e.g., via determination of prothrombin time (PT), partial thromboplastin time (PTT), and international normalized ratio (INR)), and/or blood counts (e.g., hemoglobin or hematocrit, white blood cell levels with differential, and platelets). In some embodiments, the system **10** and/or the sensing element **302** may be configured to detect and/or measure circulating tumor cells, circulating tumor DNA, circulating RNA, multigene sequencing of germ line or tumor DNA, markers of inflammation such as cytokines, C reactive protein, erythrocyte sedimentation rate, tumor markers (PSA, beta-HCG, AFP, LDH, CA 125, CA 19-9, CEA, etc.), seizure activity (procalcitonin), and others.

[0091] As previously mentioned, the system **10** may determine one or more physiological parameters based on data obtained by the sensing element **302** and/or one or more other physiological parameter(s). For example, the system **10** may be configured to determine physiological parameters such as heart rate, temperature, blood pressure (e.g., systolic blood pressure, diastolic blood pressure, mean arterial blood pressure), cardiac output, ejection fraction, pulmonary artery pressure, pulmonary capillary wedge pressure, left atrial pressure, blood flow rate, blood velocity, pulse wave speed, volumetric flow rate, reflected pressure wave amplitude, augmentation index, flow reserve, resistance reserve, resistive index, capacitance reserve, hematocrit, heart rhythm, electrocardiogram (ECG) tracings, body fat percentage, activity level, body movement, falls, gait analysis, seizure activity, blood glucose levels, drug/medication levels, blood gas constituents and blood gas levels (e.g., oxygen, carbon dioxide, etc.), lactate levels, hormone levels (such as cortisol, thyroid hormone (T4, T3, free T4, free T3), TSH, ACTH, parathyroid hormone), medication concentration in the blood, pharmacokinetic and pharmacodynamic data, and/or any correlates and/or derivatives of the foregoing measurements and parameters (e.g., raw data values, including voltages and/or other directly measured values). In some embodiments, data obtained by the sensing element **302** can be utilized or characterized as a physiological parameter without any additional processing by the system **10**.

[0092] The system **10** may also determine and/or monitor derivatives of any of the foregoing parameters (e.g., physiological parameters, device performance parameters, treatment parameters, identity parameters, etc.), such as a rate of change of a particular parameter, a change in a particular parameter over a particular time frame, etc. As but a few examples, the system **10** may be configured to determine a temperature over a specified time, a maximum temperature, a maximum average temperature, a minimum temperature, a temperature at a predetermined or calculated time relative to a predetermined or calculated temperature, an average temperature over a specified time, a maximum blood flow, a minimum blood flow, a blood flow at a predetermined or calculated time relative to a predetermined or calculated blood flow, an average blood flow over time, a maximum impedance, a minimum impedance, an impedance at a predetermined or calculated time relative to a predetermined or calculated impedance, a change in impedance over a specified time, a change in impedance relative to a change in temperature over a specified time, a change in heart rate over time, a change in respiratory rate over time, activity level over a specified time and/or at a specified time of day, and other suitable derivatives.

[0093] Data may be obtained continuously or periodically at one or more predetermined times, ranges of times, calculated times, and/or times when or relative to when a measured event occurs. Likewise, parameters (physiological or otherwise) may be determined continuously or periodically at one or more predetermined times, ranges of times, calculated times, and/or times when or relative to when a measured event occurs.

[0094] Based on the determined parameters, the system **10** of the present technology can be configured to provide an indication of the patient's health, the performance and/or health of device, and/or the status of a treatment to the patient and/or a clinician. For example, the controller may compare one or more physiological parameters to a predetermined threshold or range and, based on the comparison, provide an indication of the patient's health. If the determined physiological parameter(s) is above or below the predetermined threshold or outside of the predetermined range, the system **10** may provide an indication that the patient is at risk of, or has already developed, a medical condition characterized by symptoms associated with the determined physiological parameters. As used herein, a “predetermined range” refers to a set range of values, and “outside of a/the predetermined range” refers to (a) a measured or calculated range of values that only partially overlap the predetermined range or do not overlap any portion of a predetermined range of values. As used herein, a “predetermined threshold” refers to a single value or range of values, and a parameter that is “outside” of “a predetermined threshold” refers to a situation where the parameter is (a) a measured or calculated value that exceeds or fails to meet a predetermined value, (b) a measured or calculated value that falls outside of a predetermined range of values, (c) a measured or calculated range of values that only partially overlaps a predetermined range of values or does not overlap any portion of a predetermined range of values, or (d) a measured or calculated range of values where none of the values overlap with a predetermined value.

[0095] Predetermined parameter thresholds and/or ranges can be empirically determined to create a look-up table. Look-up table values can be empirically determined, for example, based on clinical studies and/or known healthy or normal values or ranges of values. The predetermined threshold may additionally or alternatively be based on a particular patient's baseline physiological parameters, a particular device's baseline performance parameters, etc. In some embodiments, the system **10** can be configured to determine the predetermined threshold and/or range based on data collected by the device to determine a patient's normal or baseline parameter or range of parameters. In some cases, a patient's baseline parameter or range of parameters may differ from known normal parameters/ranges of parameters. For example, a patient with anemia may have a lower baseline hemoglobin level, hematocrit, or red blood cell count than a non-anemic patient. Accordingly, a predetermined threshold for hemoglobin that indicates that the patient's hemoglobin is abnormal and/or indicative of a medical condition may be lower for the anemic patient than for a non-anemic patient.

[0096] In some embodiments, the system **10** may be configured to detect a pattern of measurements (of a single parameter or a combination of parameters) indicative of a health condition. In such embodiments, the individual measurements may not fall outside of a given “normal” range yet, when considered together, can still indicate a change in health status. For example, the controller may be configured to identify patterns of change in temperature, heart rate, and activity that are associated with infection, even if all three are still within a “normal” range.

[0097] Medical conditions detected and/or indicated by the system **10** may include, for example, sepsis, pulmonary embolism, metastatic spinal cord compression, anemia, dehydration/volume depletion, vomiting, pneumonia, congestive heart failure, performance status, arrythmia, neutropenic fever, acute myocardial infarction, pain, opioid toxicity, nicotine or other drug addiction or dependency, hyperglycemic/diabetic ketoacidosis, hypoglycemia, hyperkalemia, hypercalcemia, hyponatremia, one or more brain metastases, superior vena cava syndrome, gastrointestinal hemorrhage, immunotherapy-induced or radiation pneumonitis, immunotherapy-induced colitis, diarrhea, cerebrovascular accident, stroke, pathological fracture, hemoptysis, hematemesis, medication-induced QT prolongation, heart block, tumor lysis syndrome, sickle cell anemia crisis, gastroparesis/cyclic vomiting syndrome, hemophilia, cystic fibrosis, chronic pain, volume overload, hyperuricemia, and/or seizure. Any of the systems and/or devices disclosed herein may be used to monitor a patient for any of the foregoing medical conditions.

[0098] The system **10** can be configured to provide notifications to a patient and/or a clinician. For example, the device **300** and/or an external computing device (e.g., a smartphone, a PC, etc.) can be configured to provide a notification to the patient if a battery carried by the electronic device **102** is low and needs to be recharged. The system **10** may be configured to provide a notification to a patient and/or clinician if it determines that the device is occluded, overheating, has lost wireless communication, is infected or colonized, or is otherwise malfunctioning. In some embodiments, the electronic device **102** includes a notification unit configured to provide notifications to the patient without the need for an external computing device. The notification unit can include a speaker, a light, a vibration element, or another means for providing audible, visual, haptic, and/or tactile notifications to the patient. As an example, the notification unit can comprise a vibration element configured to vibrate if the patient's blood glucose has fallen outside of predetermined healthy or normal range to indicate to the patient that intervention should be taken to regulate their blood glucose.

[0099] In some embodiments, one or more parameters of a notification provided by a notification unit can be based on a type of information to be communicated to a patient. For example, a high frequency, high amplitude vibration can communicate to the patient that their blood pressure is higher than a predetermined threshold, while a low frequency, low amplitude vibration can communicate to the patient that their blood pressure is lower than a predetermined threshold. Further, various notification modalities and/or parameters can be used alone or in combination to communicate specific information to the patient. For example, the notification unit can include an LED light and a speaker. If the system **10** determines that the patient is experiencing ventricular fibrillation, the LED light can emit red light and the speaker can emit an audible notification instructing listeners to call an ambulance. In some embodiments, the LED light can emit a light of a specific color to indicate that the electronic device **102** needs to be recharged.

[0100] According to various embodiments of the present technology, the device **300** can comprise a voice recognition unit configured to obtain audio data. The voice recognition unit can include a microphone, such as any of the microphones described herein, and a controller communicatively coupled to the microphone. In some embodiments, the voice recognition unit or one or more portions thereof is communicatively coupled to the sensing element **302** or another electronic component carried by the device **300**. In some embodiments, the voice recognition unit comprises a microphone communicatively coupled to a separate controller carried by device **300** and/or an external controller.

[0101] The voice recognition unit can be configured to obtain audio data and, in some embodiments, transmit the audio data to the controller **304**. In some embodiments, the controller **304** can be configured to cause the device **300** to perform a function or action, based at least in part on the audio data. For example, a person (e.g., a patient, a clinician, etc.) can provide an audible instruction for the device **300** to obtain data characterizing a temperature of the person. Providing the audible instruction can include saying out loud “Device, take the patient's temperature.” The voice recognition unit can obtain audio data characterizing the sound waves produced by the person providing the audible instruction. The audio data can be transmitted to a controller, which can process the audio data and, based on the audio data, cause the sensing element **302** carried by the device **300** to obtain data characterizing a temperature of the patient. In some embodiments, the controller can process the temperature data and/or cause the notification unit to provide a notification communicating the patient's temperature. In another example, a person can provide an audible instruction for the device **300** to turn on, wake up, exit a lower-power mode, or otherwise activate.

[0102] According to some methods of use, the vascular access device **200** can be implanted through an incision at a first location, and the electronic device **300** can be implanted through the incision at a second location that is different than the first location. For example, an incision can be made in the upper chest region to create a subcutaneous pocket. The vascular access device **200** can be delivered through the incision to the subcutaneous pocket, and then the electronic device **300** can be separately delivered through the same incision to the subcutaneous pocket. Alternatively, the electronic device **300** can be delivered through the incision to the subcutaneous pocket, and then the vascular access device **200** can be separately delivered through the same incision to the subcutaneous pocket.

[0103] In some embodiments, the vascular access device **200** can be free-floating relative to the electronic device **300**. Alternatively, the vascular access device **200** and electronic device **300** can be mechanically coupled to one another, for example by joining them together directly or via a coupler. According to some methods of use, the vascular access device **200** and electronic device **300** are coupled to one another prior to implantation (i.e., ex vivo). According to some methods of use, the vascular access device **200** and the electronic device **300** are coupled to one another in vivo.

[0104] FIGS. **4**A-**9**B depict various coupler embodiments configured in accordance with the present technology. The couplers shown in FIGS. **4**A-**9** can be used with any of the vascular access devices **200** and electronic devices **300** disclosed herein.

[0105] FIGS. **4**A, **4**B, and **4**C, for example, show a coupler **400** and associated implantable system. FIG. **4**B shows the coupler **400** prior to engagement with the vascular access device **200** and electronic device **300**. The coupler **400** can comprise a flexible housing having a first receiving portion **402** configured to receive the vascular access device **200** and a second receiving portion **404** configured to receive the electronic device **300**. The housing may further include an intermediate portion **406** between the first and second receiving portions **402**, **404**. The intermediate portion **406** can be sufficiently flexible to allow some movement between the first and second receiving portions **402**, **404**. For example, the housing can be configured to flex and/or bend at the intermediate portion **406** so that the system **100** can better conform to the curvature of the patient's chest (or other implantation location), thereby improving patient comfort. In some embodiments, the intermediate portion **406** can be relatively thin to encourage preferential bending at the intermediate portion **406**. Alternatively, the intermediate portion **406** may be rigid such that the devices **200** and **300** are held in a desired spatial relationship. In some embodiments, the intermediate portion **406** can comprise a material that is sufficiently soft and/or thin such that the coupler **400** can be easily cut in vivo, at the intermediate portion **406**, to separate the devices **200** and **300**. Such a detachable arrangement may be beneficial should the need arise to remove only one of the devices **200** and **300** from the system.

[0106] The first receiving portion **402** can comprise a sidewall **403** that defines an opening **412** and at least partially outlines a shape complementary to that of the vascular access device **200**. For example, in some embodiments the vascular access device **200** has a circular shape, and the sidewall of the first receiving portion **402** circumscribes a portion of the circular shape. The first receiving portion **402** can have a resting diameter that is slightly less than the diameter of the vascular access device **200**, and the sidewall can be configured to stretch to accommodate the larger vascular access device **200**. The sidewall can be sufficiently elastic such that, in its deformed state, it exerts enough compressive force on the vascular access device **200** to secure the vascular access device **200** thereto. In some embodiments, the first receiving portion **402** is configured to receive and be secured to the vascular access device **200** in a snap fit arrangement.

[0107] In some embodiments, the first receiving portion **402** can have a gap **410** in the sidewall. As indicated by FIGS. **4**A and **4**B together, and as shown in the assembled side view of FIG. **4**C, the hub **202** of the vascular access device **200** can be configured to be positioned in the first receiving portion **402** such that the catheter **600** extends through the gap **410** in the sidewall. The gap **410** can have an arc length that allows the vascular access device **200** to rotate within the first receiving portion **402** so that the catheter **600** can be positioned at a variety of angles to accommodate the unique anatomical environments presented by different patients. In some embodiments, the arc length allows the vascular access device **200** to rotate between 10 and 180 degrees.

[0108] The second receiving portion **404** can comprise a sidewall **405** that defines an opening **408** and at least partially outlines a shape complementary to that of the electronic device **300**. For example, in some embodiments the electronic device **300** has an elongated shape, and the sidewall of the second receiving portion **404** circumscribes the entire elongated shape. The second receiving portion **404** can have a resting diameter that is slightly less than the diameter of the electronic device **300**, and the sidewall can be configured to stretch to accommodate the larger electronic device **300**. The sidewall can be sufficiently elastic such that, in its deformed state, it exerts enough compressive force on the electronic device **300** to secure the electronic device **300** thereto. In some embodiments, the second receiving portion **404** is configured to receive and be secured to the electronic device **300** in a snap fit arrangement.

[0109] In some embodiments, for example as shown in FIGS. **4**B and **4**C, the first and second receiving portions **402**, **404** only comprise a respective sidewall **403**, **405** and do not include any portion that extends across all or a portion of the top or bottom sides of respective openings **412**, **408**. As such, when the vascular access device **200** and electronic device **300** are positioned in the first and second receiving portions **402**, **404**, respectively, the top and bottom surfaces of the vascular access device **200** and electronic device **300** are exposed. In some embodiments, the first receiving portion **402** can comprise a bottom support (not shown) that covers all or a portion of the opening **412** at the bottom side of the sidewall **403**. The top side of the first receiving portion **402** may remain open, however, so that the reservoir can still be accessed by a needle. Additionally or alternatively, the second receiving portion **404** can comprise a bottom support (not shown) that covers all or a portion of the opening **408** at the bottom side of the sidewall **405**. Additionally or alternatively, the second receiving portion **404** can comprise a top support (not shown) that covers all or a portion of the opening **408** at the top side of the sidewall **405**. In those embodiments in which the second receiving portion **404** has both top and bottom supports, the sidewall **405** can have a gap (similar to gap **410**) to allow the device **300** to be inserted into the opening **408**. In any of the foregoing embodiments, any portion of the first and/or second receiving portions **402**, **404** (e.g., the sidewall, the top support, the bottom support, etc.) can have apertures extending therethrough that enable contact between electrodes and/or other sensing elements of the respective device **200**, **300** and adjacent body tissue.

[0110] As shown in FIG. **4**B, the gap **410** can be positioned along a side of the first receiving portion **402** that is laterally opposite of the intermediate portion **406** and/or second receiving portion **404**. As such, the catheter **600** is configured to extend laterally away from the intermediate portion **406** and/or second receiving portion **404**. In other embodiments, the gap **410** can be positioned elsewhere along the first receiving portion **402**. For example, as shown in FIGS. **5**A and **5**B, the gap **410** can be positioned along the first receiving portion **402** such that, when the vascular access device **200** is positioned within the first receiving portion **402**, the catheter **600** extends substantially perpendicular to the longitudinal axis of the second receiving portion **404** (plus or minus 5-10 degrees). As shown in FIGS. **6**A and **6**B, in some embodiments the gap **410** is

positioned along the first receiving portion **402** such that, when the vascular access device **200** is positioned within the first receiving portion **402**, the catheter **600** extends laterally away from the second receiving portion **404** and forms an angle of about 135 degrees with the longitudinal axis of the second receiving portion **404** (plus or minus 5-10 degrees). As shown in FIGS. **7**A and **7**B, in some embodiments the gap **410** is positioned along the first receiving portion **402** such that, when the vascular access device **200** is positioned within the first receiving portion **402**, the catheter **600** extends laterally towards the second receiving portion **404** and forms an angle of about 45 degrees with the longitudinal axis of the second receiving portion **404** (plus or minus 5-10 degrees).
[0111] For any of the foregoing embodiments, it will be appreciated that the gap in the first receiving portion can be positioned at any location along the circumference and/or curved boundary defining the first receiving portion. Likewise, the gap can have any width that is sufficient to receive the vascular access device **200** therethrough.
[0112] As shown in FIGS. **8** and **9**, in some embodiments the housing of the coupler **400** can have a bottom support **800** and a lip **802** that extends radially inwardly from a top portion of the sidewall **804** defining the first receiving portion **402**. The lip **802** can be configured to extend over a portion of the vascular access device **200** positioned therein, thereby limiting and/or preventing vertical movement of the vascular access device **200** relative to the coupler **400**. The lip **802** can extend around the entire perimeter of the sidewall **804** of the first receiving portion **402**, or may extend around less than the entire perimeter of the sidewall **804**. To secure the vascular access device **200** to the first receiving portion **402**, the vascular access device **200** can be introduced through a top opening in the first receiving portion **402**, thereby initially forcing the lip **802** to bend downwardly. Once the vascular access device **200** is positioned against the bottom support **800**, the lip **802** can be free to return to its resting position, which extends over a portion of the vascular access device **200**. Additionally or alternatively, the second receiving portion **404** can have a bottom surface and a lip extending radially inwardly along all or a portion of its sidewall.
[0113] As shown in FIG. **9**, in some embodiments the first receiving portion **402** can comprise a bottom support **900** that spans less than the entire area defined by the sidewalls of the first receiving portion **402**. In such embodiments, the first receiving portion **402** can have a first sidewall **902** closest to the intermediate portion **406** and a second sidewall **904** laterally opposite of the first sidewall **902** and connected to the first sidewall **902** via the bottom support **900**. The first receiving portion **402** can further include first and second gaps **906**, **908** between the first and second sidewalls **902**, **904**. The end portions of the sidewalls **902**, **904** (e.g., closest to the gaps **906**, **908**) can be configured to bend in a radial direction to facilitate insertion of the vascular access device **200** through the side gaps **906**, **908**. For example, when inserting the vascular access device **200**, the ends may bend radially inwardly, then spring back to their resting positions when the vascular access device **200** is fully inserted. Likewise, when the vascular access device **200** is removed through one of the side gaps **906**, **908**, the end portions can bend radially outwardly to allow passage of the vascular access device **200**, then return to their resting positions when the vascular access device **200** is removed. Additionally or alternatively, the bottom support **900** can be configured to bend to increase the distance between adjacent end portions and enable insertion and removal of the vascular access device **200**.
[0114] When implanting the vascular access device **200**, the hub **202** can be implanted beneath the patient's skin, and the catheter **220** can be inserted into a targeted cardiovascular location, such as a targeted blood vessel. The blood vessel, for example, can be an artery or a vein, such as the internal jugular vein or the subclavian vein. In operation, a clinician inserts a needle (e.g., a non-coring or Huber-type needle) through the skin, through the self-sealing septum **212**, and into the fluid reservoir **210**. To introduce fluid (e.g., medication, contrast agent, saline solution, etc.) into the patient's blood vessel, the clinician may inject the fluid through the needle N, which then flows through the reservoir **210**, the catheter **220**, and into the vessel. In some cases, the physician may inject fluid through the needle to fill the reservoir **210** for postponed delivery into the vessel. To

remove fluid from the vessel (e.g., to aspirate blood from the vessel for testing), the clinician can apply suction via the needle, thereby withdrawing fluid (e.g., blood) from the vessel into the catheter **220**, into the fluid reservoir **210**, and into the needle. After aspiration or fluid delivery, the device **200** can be flushed and/or locked. A flushing fluid (e.g., saline solution, heparin solution, etc.) can be injected through the needle, into the reservoir **210**, and out of the catheter **220** such that the flushing fluid transports undesired material (e.g., residual medication, debris, etc.) out of the device **200**. In some embodiments, the device **200** can be locked by delivering a volume of a locking fluid (e.g., a solution carrying at least one of sodium chloride, an anticoagulant agent, a thrombolytic agent, an antimicrobial agent, an antiseptic agent, or another suitable agent) to the catheter **220** such that the locking fluid remains in the catheter lumen and prevents or limits ingress of blood into the catheter lumen or occlusion of the catheter lumen. When the procedure is completed, the clinician removes the needle, the self-sealing septum **212** resumes a closed configuration, and the device **200** may remain in place beneath the patient's skin.

[0115] FIGS. **10**A-**10**D illustrate several variations on a coupler **1000** comprising one or more recesses and/or openings in the sidewall **1003** of the first receiving portion **1002** to accommodate the catheter **600** of the vascular access device **200**. As shown in FIG. **10**A, in some embodiments the coupler **1000** comprises a recess **1022** extending downwardly from a top side **1003***a* of the sidewall **1003** of the first receiving portion **1002**. Aside from the gap created by the recess **1022**, the sidewall **1003** can be configured to extend fully around a boundary of the vascular access device **200**. To secure the device **200** to the first receiving portion **1002**, a user can insert the device **200** downwardly through a top side of the opening **412** until the catheter **600** sits within a portion of the sidewall **1003** defining the recess **1022**. In some embodiments, the device **200** can first be positioned in the first receiving portion **1002** without the catheter **600**, and then the catheter **600** can be coupled to the device **200**.

[0116] As shown in FIG. **10**B, in some embodiments the coupler **1000** comprises a recess **1024** extending upwardly from a bottom side **1003***b* of the sidewall **1003** of the first receiving portion **1002**. Aside from the gap created by the recess **1024**, the sidewall **1003** can be configured to extend fully around a boundary of the vascular access device **200**. To secure the device **200** to the first receiving portion **1002**, a user can insert the device **200** upwardly through a bottom side of the opening **412** until the catheter **600** sits within a portion of the sidewall **1003** defining the recess **1022**. In some embodiments, the device **200** can first be positioned in the first receiving portion **1002** without the catheter **600**, and then the catheter **600** can be coupled to the device **200**.

[0117] As shown in FIG. **10**C, in some embodiments the coupler **1000** comprises an opening **1026** extending through the sidewall **1003** of the receiving portion **1002** at a location between the top and bottom sides of the sidewall. Aside from the gap created by the opening **1026**, the sidewall **1003** can be configured to extend fully around a boundary of the vascular access device **200**. To secure the device **200** to the first receiving portion **1002**, a user can guide the catheter **600** through the opening **1026** as the device **200** is inserted into the opening **412** (through the top or bottom side). In some embodiments, the device **200** can first be positioned in the first receiving portion **1002** without the catheter **600**, and then the catheter **600** can be coupled to the device **200**.

[0118] In some embodiments, the coupler **1000** can have more than one top or bottom recess (e.g., three recesses, four recesses, etc.). For example, as shown in FIG. **10**D, the coupler **1000** can have a first recess **1022** and a second recess **1028**, each at a different location along a boundary of the first receiving portion **402**. This way, the first receiving portion **402** can receive the vascular access device **200** in different orientations and/or can receive a vascular access device with multiple catheters extending therefrom. Similarly, the coupler **1000** can have more than one opening (not shown). In some embodiments, the coupler **1000** has one or more top recesses **1022**, one or more bottom recesses **1024**, and/or one or more openings **1026** (e.g., the coupler **1000** can have one top recess **1022** and two bottom recesses **1024**; a top recess **1022**, a bottom recess **1024**, and an opening **1026**; three bottom recesses **1024** and two openings **1026**, etc.). Moreover, in any of the foregoing embodiments, the coupler **1000** can comprise an elastic or otherwise deformable material that can stretch to receive the vascular access device **200** and/or electronic device **300** therein and secures the respective devices via tension. In these and other embodiments the coupler **1000** can be configured to engage the vascular access device **200** and/or electronic device **300** in a snap fit arrangement.

[0119] According to various aspects of the technology, the coupler may be integrated with the housing of the electronic device **300**. For example, as shown in FIGS. **11**A-**11**C, the electronic device **300** can comprise a housing **1100** (such as an hermetically sealed enclosure) that defines a recessed portion **1102** configured to receive at least a portion of the vascular access device **200**. The recessed portion **1102** can have a shape corresponding to a footprint of all or some of the vascular access device **200** such that the vascular access device **200** is configured to be secured to the housing **1100** by a friction and/or snap fit arrangement. As shown in FIG. **11**A, the recessed portion **1102** can be oriented such that, when the vascular access device **200** is received within the recessed portion **1102**, a longitudinal axis L**2** of the vascular access device **200** is angled with respect to the longitudinal axis L**3** of the electronic device **300** and/or housing **1100**. In some embodiments, the recessed portion **1102** can be oriented such that, when the vascular access device **200** is received within the recessed portion **1102**, the longitudinal axis L**2** of the vascular access device **200** is substantially perpendicular to the longitudinal axis L**3** of the electronic device **300** and/or housing **1100**. In certain embodiments, the recessed portion **1102** can be oriented such that, when the vascular access device **200** is received within the recessed portion **1102**, the longitudinal axis L**2** of the vascular access device **200** is substantially aligned with and/or parallel to the longitudinal axis L**3** of the electronic device **300** and/or housing **1100** (for example, as shown in FIGS. **13**A and **13**B).

[0120] According to several aspects of the technology, the housing **1100** may include one or more engagement members configured to further engage and secure the vascular access device **200** while the vascular access device **200** is positioned in and/or on the recessed portion **1102**. For example, as shown in FIG. **11**B, the housing **1100** can include one or more protrusions **1104** configured to engage a portion of the vascular access device **200** that is not positioned within the recessed portion **1102**. The protrusion(s) **1104** can surround all or a portion of the vascular access device **200**. In some embodiments, for example as shown in FIG. **11**C, the housing **1100** can include one or more posts **1106** positioned within the recessed portion **1101** and configured to engage a bottom surface of the vascular access device **200**. The vascular access device **200** can have one or more recesses configured to receive the posts **1106**. In some embodiments, the posts **1106** can be configured to be received within suture holes of the vascular access device **200**.

[0121] FIG. **12**A shows an electronic device **300** having a housing **1200** similar to housing **1100**, except in FIG. **12**A, only one side of the recessed portion **1202** is vertically bound by a sidewall of the housing **1200**. In several of such embodiments, the housing **1200** can include one or more engagement members to secure the vascular access device **200** to the housing **1200**. The engagement members can include a laterally extending protrusion **1204** (shown in FIGS. **12**B and **12**C), one or more posts (not shown), and/or other features.

[0122] FIG. **13**A shows an electronic device **300** having a housing **1300** similar to housing **1200**, except in FIG. **13**A, the recessed portion **1302** is oriented such that, when the vascular access device **200** is received within the recessed portion **1302**, the longitudinal axis L**2** of the vascular access device **200** is substantially aligned with and/or parallel to the longitudinal axis L**3** of the electronic device **300** and/or housing **1300**. FIG. **13**B shows a variation of the housing **1300** in which the portion of the housing **1300** forming the vertical sidewalls that define the recessed portion **1302** are configured to surround all but the catheter **600** (and adjacent supporting structure) when the vascular access device **200** is positioned in the recessed portion **1302**. In FIG. **13**A, the vertical sidewalls surround only a portion of the perimeter of the vascular access device **200**. The embodiments shown and described with respect to FIGS. **13**A and **13**B can include one or more

engagement members described herein.
[0123] FIG. **14** shows an electronic device **300** having a housing **1400** comprising one or more openings through which a portion of the vascular access device **200** may protrude. For example, the housing **1400** can include a first opening **1404** configured to receive a top portion of the reservoir and/or septum therethrough, and a second opening **1406** configured to receive the catheter **600** therethrough. In some embodiments the housing **1400** comprises a single opening or more than two openings (e.g., three openings, four openings, etc.). The remainder of the housing **1400** can be surround and constrain the vascular access device **200**. In some embodiments, the electronic device **300** optionally includes an extension **1408** comprising an electrode. While in FIG. **14** the openings **1404**, **1406** are disposed proximate an end portion of the electronic device **300**, in some embodiments the openings can be disposed at an intermediate portion of the electronic device **300**. For example, FIGS. **15**A and **15**B show an electronic device **300** having a housing **1500** with first and second openings **1502**, **1504** disposed at an intermediate region (e.g., away from the longitudinal ends) of the electronic device **300** with end portions **300***a*, **300***b* of the housing **1500** on either side.
[0124] FIGS. **16**A-**16**C show a housing **1600** similar to housing **1500**. The housing **1600** has a recessed portion **1602** (best shown in FIG. **16**C) configured to receive the vascular access device **200** therein. The housing **1600** can have one or more openings configured to receive one or more portions of the vascular access device **200** therethrough. For example, as shown in FIGS. **16**A and **16**B, the housing **1600** can include a side opening **1606** extending therethrough that is continuous with the recessed portion **1602** and configured to receive the catheter **600** (and/or adjacent supporting structure) therethrough. Additionally or alternatively, the housing **1600** can include a top opening **1608** extending therethrough that is continuous with the recessed portion **1602** a portion of the vascular access device **200**, such as a top portion of the reservoir and/or septum, therethrough. In some embodiments, the housing **1600** includes one or more engagement members configured to further secure the vascular access device **200** to the housing **1600**. For example, as best shown in FIG. **16**C, in some embodiments a surface of the housing **1600** that defines the recessed portion **1602** can include one or more posts **1604** (only one labeled) extending therefrom and configured to mate with recesses **1612** (only one labeled in FIG. **17**A) at a top portion of the vascular access device **200**. In addition to or instead of the posts **1604**, the housing **1600** can include other engagement members. In some embodiments, the housing **1600** can optionally include an electrode **1610** disposed on one or both sides of the recessed portion **1602**.
[0125] In some embodiments, the system can further comprise a plug that is configured to be disposed within the recessed portion and/or openings in the housing of the electronic device **300** when the vascular access device **200** is not present. An example of a plug **1700** and its use is depicted in FIGS. **17**A-**17**C. In some embodiments, the electronic device **300** may be implanted without the vascular access device **200** and without the plug **1700**. In some embodiments, the electronic device **300** may be implanted only with the plug **1700**. In some embodiments, the electronic device **300** may be implanted with the vascular access device **200** coupled thereto, and the vascular access device **200** can be removed at a later date and replaced by the plug **1700**.
[0126] FIG. **18** shows an electronic device **300** comprising a housing **1800** that is similar to the housing **1100** shown in FIG. **11**A, except in FIG. **18**, the recessed portion **1802** is positioned at an intermediate portion of the housing **1800**. The embodiments shown and described with respect to FIG. **18** can include one or more engagement members described herein.
[0127] FIGS. **19**A and **19**B show an electronic device **300** comprising a housing **1900** that is similar to the housing **1100** shown in FIG. **11**A, except in FIGS. **19**A and **19**B, the housing **1900** includes multiple openings **1904***a*, **1904***b*, and **1904***c*, each configured to receive the catheter **600** (and/or adjacent supporting structure) therethrough. The recessed portion **1902** is further shaped to accommodate the vascular access device **200** in different orientations, such as with a longitudinal axis of the vascular access device **200** substantially aligned with and/or parallel to the longitudinal

axis of the housing **1900** (as shown in FIG. **19**A), with the longitudinal axis of the vascular access device **200** substantially perpendicular to the longitudinal axis of the housing **1900**, and/or other orientations. The shape of the recessed portion **1902** allows the operator to place the vascular access device **200** in more than one possible configuration in a subcutaneous pocket based on a patient's unique anatomy and body habitus. The embodiments shown and described with respect to FIGS. **19**A and **19**B can include one or more engagement members described herein.

CONCLUSION

[0128] Although many of the embodiments are described above with respect to vascular access devices, the technology is applicable to other applications and/or other approaches, such as other types of implantable medical devices (e.g., pacemakers, implantable cardioverter/defibrillators (ICD), deep brain stimulators, insulin pumps, infusion ports, orthopedic devices, and monitoring devices such as pulmonary artery pressure monitors). Moreover, other embodiments in addition to those described herein are within the scope of the technology. Additionally, several other embodiments of the technology can have different configurations, components, or procedures than those described herein. A person of ordinary skill in the art, therefore, will accordingly understand that the technology can have other embodiments with additional elements, or the technology can have other embodiments without several of the features shown and described above with reference to FIGS. **1-19**B.

[0129] The descriptions of embodiments of the technology are not intended to be exhaustive or to limit the technology to the precise form disclosed above. Where the context permits, singular or plural terms may also include the plural or singular term, respectively. Although specific embodiments of, and examples for, the technology are described above for illustrative purposes, various equivalent modifications are possible within the scope of the technology, as those skilled in the relevant art will recognize. For example, while steps are presented in a given order, alternative embodiments may perform steps in a different order. The various embodiments described herein may also be combined to provide further embodiments.

[0130] As used herein, the terms “generally,” “substantially,” “about,” and similar terms are used as terms of approximation and not as terms of degree, and are intended to account for the inherent variations in measured or calculated values that would be recognized by those of ordinary skill in the art.

[0131] Moreover, unless the word “or” is expressly limited to mean only a single item exclusive from the other items in reference to a list of two or more items, then the use of “or” in such a list is to be interpreted as including (a) any single item in the list, (b) all of the items in the list, or (c) any combination of the items in the list. Additionally, the term “comprising” is used throughout to mean including at least the recited feature(s) such that any greater number of the same feature and/or additional types of other features are not precluded. It will also be appreciated that specific embodiments have been described herein for purposes of illustration, but that various modifications may be made without deviating from the technology. Further, while advantages associated with certain embodiments of the technology have been described in the context of those embodiments, other embodiments may also exhibit such advantages, and not all embodiments need necessarily exhibit such advantages to fall within the scope of the technology. Accordingly, the disclosure and associated technology can encompass other embodiments not expressly shown or described herein.

[0132] For the purposes of this specification and appended claims, unless otherwise indicated, all numerical values used in the specification and claims, are to be understood as being modified in all instances by the term “about.” Accordingly, unless indicated to the contrary, the numerical parameters set forth in the following specification and attached claims are approximations that may vary depending upon the desired properties sought to be obtained by the present invention. At the very least, and not as an attempt to limit the application of the doctrine of equivalents to the scope of the claims, each numerical parameter should at least be construed in light of the number of reported significant digits and by applying ordinary rounding techniques.

[0133] Notwithstanding that the numerical ranges and parameters setting forth the broad scope of the invention are approximations, the numerical values set forth in the specific examples are reported as precisely as possible. Any numerical value, however, inherently contains certain errors necessarily resulting from the standard deviation found in their respective testing measurements. Moreover, all ranges disclosed herein are to be understood to encompass any and all subranges subsumed therein. For example, a range of “1 to 10” includes any and all subranges between (and including) the minimum value of 1 and the maximum value of 10, that is, any and all subranges having a minimum value of equal to or greater than 1 and a maximum value of equal to or less than 10, e.g., 5.5 to 10.

Claims

1. An implantable vascular access system, comprising: a vascular access device comprising: a reservoir body defining an interior chamber configured to receive fluid therein; a self-sealing septum covering the interior chamber; an outlet port in fluid communication with the chamber; a catheter configured to be mated to the outlet port of the vascular access device; an electronic device including one or more sensing elements configured to obtain physiological measurements; and a coupler configured to at least partially engage the vascular access device and the electronic device, the coupler comprising: a flexible housing having a first receiving portion configured to receive the vascular access device, a second receiving portion configured to receive the electronic device, an intermediate portion between the first receiving portion and the second receiving portion.
2. The implantable vascular access system of claim 1, wherein the intermediate portion between the first receiving portion and the second receiving portion has a flexibility configured to facilitate bending of the intermediate portion.
3. The implantable vascular access system of claim 1, wherein the intermediate portion is defined by a rigid material configured such that a fixed spacing is maintained between the first receiving portion and the second receiving portion.
4. The implantable vascular access system of claim 1, wherein the first receiving portion is defined by a sidewall and the sidewall includes at least one gap for receiving at least a portion of the catheter.
5. The implantable vascular access system of claim 4, wherein the at least one gap in the sidewall of the first receiving portion is arranged laterally opposite the intermediate portion of the coupler.
6. The implantable vascular access system of claim 4, wherein the second receiving portion is defined by a first longitudinal axis and wherein the at least one gap in the sidewall is arranged such that the catheter extends along a second longitudinal axis, such that an angle is defined between the first longitudinal axis and the second longitudinal axis.
7. The implantable vascular access system of claim 6, wherein the angle is approximately 90 degrees.
8. The implantable vascular access system of claim 6, wherein the angle is approximately 135 degrees.
9. The implantable vascular access system of claim 1, wherein the second receiving portion includes a sidewall that defines an opening complimentary in shape to the electronic device.
10. The implantable vascular access system of claim 1, wherein the first receiving portion is defined by a sidewall and a lip extends radially inwardly from a top surface of the sidewall.
11. A method for remotely monitoring the health of a patient, the method comprising: implanting an electronic device; and implanting a vascular access device after implanting the electronic device; and after implanting the vascular access device, securing the vascular access device and the electronic device to a coupler, the coupler defined by a flexible housing having at least a first receiving portion for receiving the vascular access device and a second receiving portion for receiving the electronic device.

12. The method of claim 11, wherein the vascular access drive includes: a hub comprising a housing defining a reservoir configured to receive a fluid therein; and a catheter comprising a proximal end portion configured to be secured to the hub, a distal end portion configured to be positioned within a body lumen of a patient, and a lumen extending therethrough, the lumen being a fluid communication with the reservoir.
13. The method of claim 11, wherein the electronic device comprises: a sensing element, the sensing element being configured to obtain data characterizing a physiological parameter of a patient.
14. The method of claim 13, wherein the electronic device comprises: a data communications module communicatively coupled to the sensing element and configured to transmit the data obtained by the sensing element to an external computing device.
15. The method of claim 12, wherein the coupler further includes an intermediate portion positioned directly between the first receiving portion and the second receiving portion.
16. The method of claim 15, wherein the intermediate portion is defined by a flexible material.
17. A method for remotely monitoring the health of a patient, the method comprising: implanting an electronic device; and implanting a vascular access device after implanting the electronic device; and after implanting the vascular access device, securing the vascular access device and the electronic device to a coupler, the coupler defined by a flexible housing having at least a first receiving portion for receiving the vascular access device and a second receiving portion for receiving the electronic device.
18. The method of claim 17, wherein the coupler further includes an intermediate portion arranged between the first receiving portion and the second receiving portion, the first receiving portion and the second receiving portion arranged laterally adjacent one another.
19. The method of claim 17, wherein the vascular access device comprises: a hub comprising a housing defining a reservoir configured to receive a fluid therein; and a catheter comprising a proximal end portion configured to be secured to the hub, a distal end portion configured to be positioned within a body lumen of a patient, and a lumen extending therethrough, the lumen being in fluid communication with the reservoir.
20. The method of claim 17, wherein the electronic device comprises: a sensing element, the sensing element being configured to obtain data characterizing a physiological parameter of a patient; and a data communications module communicatively coupled to the sensing element and configured to transmit the data obtained by the sensing element to an external computing device.